

Risk Timing and Regime Awareness in Hybrid Machine Learning-Based Investment Strategies

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Abstract

Financial markets are characterized by high dimensionality, regime dependence, and time-varying risk, conditions under which traditional linear and static investment models often struggle. In this study we develop a hybrid machine-learning framework that integrates regularized supervised learning, dimensionality reduction, and unsupervised regime identification to address these challenges.

The framework combines Elastic Net regression for stress-signal prediction, Principal Component Analysis (PCA) for factor extraction and noise reduction, and K-Means clustering for regime detection. These components are implemented within a disciplined time-series validation structure that includes rolling cross-validation, walk-forward evaluation, and strict out-of-sample backtesting. We prioritize economic interpretability, stability, and implementability rather than maximizing in-sample performance.

Empirical results using multi-asset data indicate that the hybrid model achieves higher risk-adjusted returns and improved drawdown control relative to a Hidden Markov Model (HMM) benchmark. While formal statistical tests do not reject equality of Sharpe ratios, the hybrid strategy exhibits persistent net asset value outperformance driven by superior risk timing rather than increased leverage or complexity.

Rather than relying on model complexity, we find that performance differences are driven primarily by how signals are regularized and validated, and by how those signals are ultimately translated into allocation decisions in practice. In this sense, the framework serves as a practical example of how machine learning can be incorporated into portfolio management without sacrificing interpretability or implementation discipline.

Keywords: *Machine Learning in Finance; Alpha Generation; Elastic Net; Principal Component Analysis; K-Means Clustering; Regime Detection; Regularization; Portfolio Allocation; Risk Management; Time-Series Cross-Validation; Walk-Forward Backtesting*

1. Introduction

Designing investment strategies in modern financial markets has become increasingly difficult as datasets grow larger, correlations become unstable, and market behavior shifts across distinct regimes. While richer data environments offer more potential signals, they also increase the likelihood of overfitting and fragile inference, particularly when standard linear or static models are applied without careful validation. In practice, many strategies that appear promising in sample fail precisely because they do not adapt well to changing risk conditions.

Machine learning methods offer tools that are well suited to this setting, especially through regularization, dimensionality reduction, and adaptive learning mechanisms. At the same time, the growing use of complex models in asset management has introduced new concerns. Many empirical studies emphasize predictive performance while paying less attention to interpretability, economic plausibility, or the time-series structure of financial data. As a result, reported gains are

often difficult to replicate or implement outside controlled backtests.

This study takes a deliberately conservative approach. Instead of introducing a new algorithm, we combine well-understood machine learning techniques into a single investment framework that emphasizes robustness and transparency. Elastic Net regression is used to generate forward-looking stress signals under correlated predictors, Principal Component Analysis (PCA) is applied to extract dominant latent factors and suppress noise, and K-Means clustering is employed to identify broad market regimes. These components are evaluated using rolling estimation and walk-forward testing to ensure that all decisions rely only on information available at the time. The goal is not to maximize short-term performance, but to examine whether a disciplined and interpretable structure can produce stable behavior across market regimes under realistic conditions.

2. Data and Preprocessing

The analysis uses daily financial market data for U.S. Treasury bonds (TLT), gold (GLD), equities (SPY), and market volatility (VIX), sourced from Yahoo Finance. These assets are widely used in empirical finance to proxy broad market risk factors and stress conditions, making them suitable inputs for regime detection and stress modeling (Gu, Kelly, and Xiu, 2020).

All price series are aligned chronologically and forward-filled to address missing observations. Log returns are then computed, and initial observations lost to differencing are removed. These preprocessing steps follow standard practice in financial time-series analysis and help ensure numerical stability and consistency in downstream modeling (Tsay, 2010).

Prior to modeling, all features are standardized to ensure comparability across variables. Principal Component Analysis (PCA) is applied to standardized inputs to extract dominant sources of variation, while Elastic Net regression explicitly relies on scaled features to produce stable and interpretable regularized estimates. Proper normalization is particularly important for regularized and distance-based methods such as PCA and K-Means clustering (Hastie, Tibshirani, and Friedman, 2009).

Model evaluation relies on time-series-aware validation rather than random train–test splits. Rolling windows and walk-forward cross-validation are used to preserve the temporal ordering of observations and prevent look-ahead bias, which is essential for realistic performance assessment in financial applications (López de Prado, 2018; Harvey, Liu, and Zhu, 2016).

3. Methodology

3.1 Elastic Net

3.1.1 Model Selection Rationale

Elastic Net regression is used to generate forward-looking stress signals under correlated predictors. Financial features often exhibit strong multicollinearity, which can lead to unstable coefficient estimates in ordinary least squares or pure LASSO models. Elastic Net mitigates this issue by combining L1 and L2 regularization, allowing for both variable selection and coefficient stabilization (Zou and Hastie, 2005). The Elastic Net objective function combines LASSO and Ridge. Given Predictors $X \in \mathbb{R}^{n \times p}$ and target variable $y \in \mathbb{R}^n$, Elastic net solved the optimization problem

$$\min_{\beta} \frac{1}{2n} \|y - X\beta\|_2^2 + \lambda \left(\alpha \|\beta\|_1 + \frac{1 - \alpha}{2} \|\beta\|_2^2 \right)$$

where λ controls the overall strength of regularization and $\alpha \in [0,1]$ determines the balance between LASSO and Ridge penalties. Throughout the implementation, predictors are standardized prior to estimation to ensure that regularization is applied uniformly across features, consistent with standard practice in regularized regression (Hastie, Tibshirani, and Friedman, 2009).

Hyperparameters are selected using time-series cross-validation rather than random splits to preserve temporal ordering and avoid look-ahead bias. The resulting stress probability serves as a forward-looking indicator that informs regime-aware allocation decisions, rather than as a standalone prediction target.

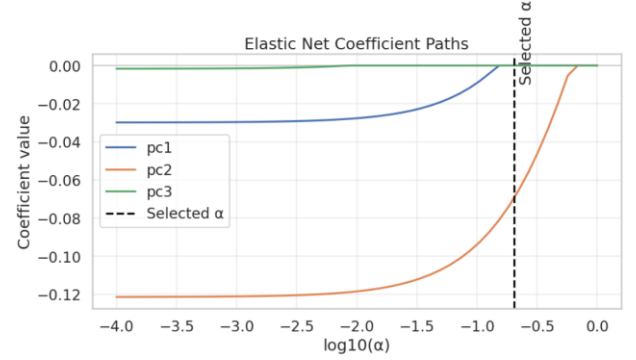


Figure 1. Elastic Net Coefficient Paths for PCA Factors

Figure 1 illustrates the coefficient paths of the Elastic Net model applied to PCA-derived factors as a function of the regularization parameter. As regularization increases, coefficients shrink smoothly toward zero rather than being eliminated abruptly. The selected cross-validated value ($\alpha \approx 0.204$) retains exposure to dominant components while suppressing weaker and less stable factors. This behavior highlights the role of disciplined regularization in improving estimation stability without relying on excessive model complexity, a property shown to be particularly important in correlated financial feature spaces (Gu, Kelly, and Xiu, 2020).



Figure 2. Elastic Net Walk-Forward In-Sample vs Out-of-Sample R^2

Figure 2 reports walk-forward in-sample and out-of-sample R^2 for the stress prediction model. The close

alignment between the two series indicates limited overfitting, despite modest and occasionally negative out-of-sample values during major regime shifts such as the Global Financial Crisis and the COVID-19 shock. This pattern is consistent with the low signal-to-noise nature of financial prediction problems and underscores the importance of evaluating models based on stability and adaptability rather than raw predictive accuracy (López de Prado, 2018).

While Elastic Net introduces bias through coefficient shrinkage, this trade-off improves robustness and interpretability in settings characterized by multicollinearity and limited effective sample sizes. By gradually shrinking unstable predictors rather than performing hard selection, the model produces smoother factor exposures across re-estimation windows, which translates into lower portfolio turnover and improved out-of-sample behavior. These properties make Elastic Net well suited for regime-aware investment strategy design under realistic market conditions (Brodie et al., 2009; Harvey, Liu, and Zhu, 2016).

3.2 K-means clustering

3.2.1 Model Selection Rationale

K-Means clustering is used to identify latent market regimes in an unsupervised manner, without requiring ex ante regime labels. This makes it well suited for financial applications where structural states are not directly observable and may evolve over time (Nystrup, Lindström, and Madsen, 2021; Atanasova and Kočenda, 2022). By grouping observations based on similarity in factor dynamics, K-Means enables regime-aware portfolio decisions that relax the assumption of a single, stable return-generating process.

Clustering is performed on PCA-transformed features rather than raw returns. This design choice reduces noise, mitigates scale effects, and improves stability by operating on orthogonal latent factors that capture dominant sources of systematic variation (Litterman and Scheinkman, 1991). Formally, K-Means partitions observations into K clusters by minimizing within-cluster variance, assigning each observation to the nearest centroid in Euclidean space (Hastie, Tibshirani, and Friedman, 2009).

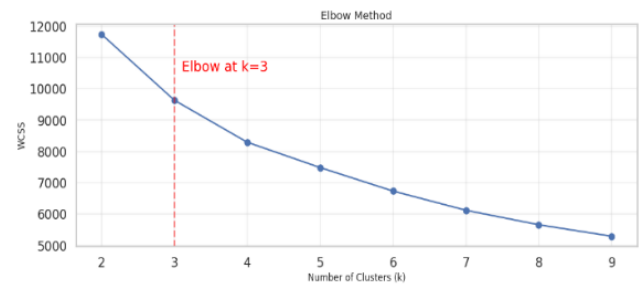


Figure 3. Elbow plot for K-Means clustering.

The number of clusters is selected using a combination of inertia-based diagnostics and economic interpretability. Figure 3 presents the elbow plot for within-cluster sum of squares, which shows a clear reduction in marginal improvement beyond $K=3$. In our experience, this choice balances flexibility with stability and yields regimes that are persistent enough to be economically meaningful.

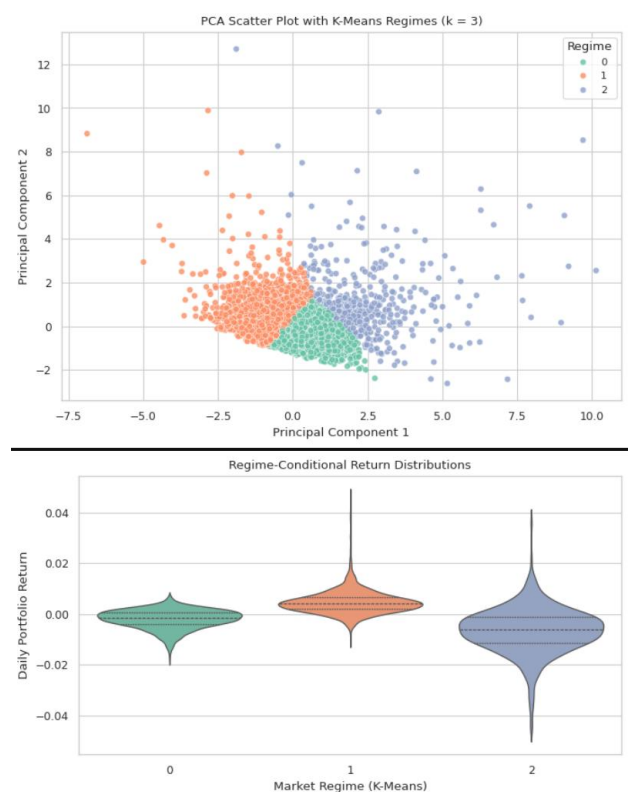


Figure 4. PCA Scatter Plot with K-Means Regimes and Regime-Conditional Return Distributions.

Figure 4 illustrates the resulting regime structure. The upper panel shows a PCA projection of the feature space with observations clustered into three regimes, while the lower panel reports regime-conditional portfolio return distributions. The identified regimes exhibit distinct return characteristics, supporting the interpretation that the clustering captures economically relevant differences in market conditions rather than purely statistical groupings.

3.3 Principal Components

3.3.1 Model Selection Rationale

Principal Component Analysis (PCA) is used to reduce dimensionality and extract latent factors that capture the dominant sources of variation in financial data. PCA has a long tradition in finance, particularly in fixed-income markets, where a small number of factors, commonly interpreted as level, slope, and curvature, explain most movements in yields and returns (Litterman and Scheinkman, 1991). More broadly, PCA is widely used to denoise correlated financial features prior to downstream modeling (Diebold and Li, 2006).

Formally, PCA decomposes the covariance (or correlation) matrix of standardized features into eigenvalues and eigenvectors, yielding orthogonal components ordered by explained variance (Hastie, Tibshirani, and Friedman, 2009). In this study, PCA is applied to standardized returns and volatility-related features to ensure that differences in scale do not distort factor extraction. Retaining a small number of components allows the model to focus on systematic variation while suppressing idiosyncratic noise.

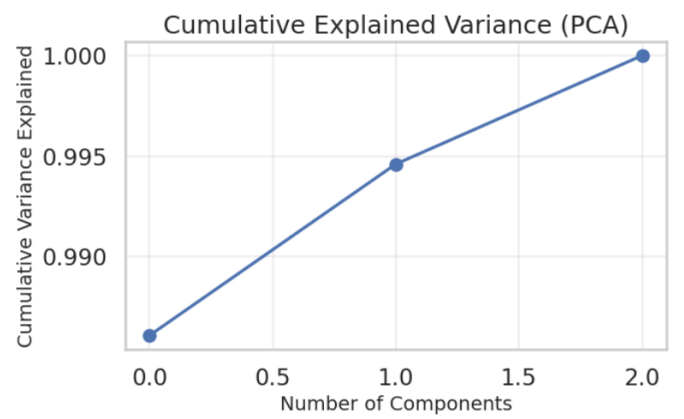


Figure 5. Cumulative Explained Variance

The number of retained components is chosen based on explained variance and economic interpretability. Figure 5 shows that the first three principal components account for the majority of variation in the feature space, indicating that the underlying structure is low-dimensional. This finding is consistent with the empirical finance literature, which documents that a small set of latent factors typically drives asset return dynamics (Litterman and Scheinkman, 1991).

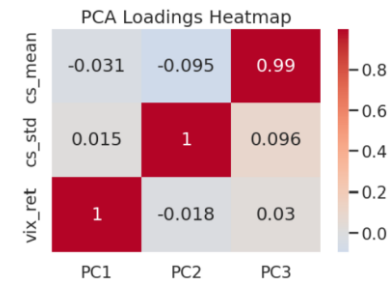


Figure 6. PCA Loadings Heatmap

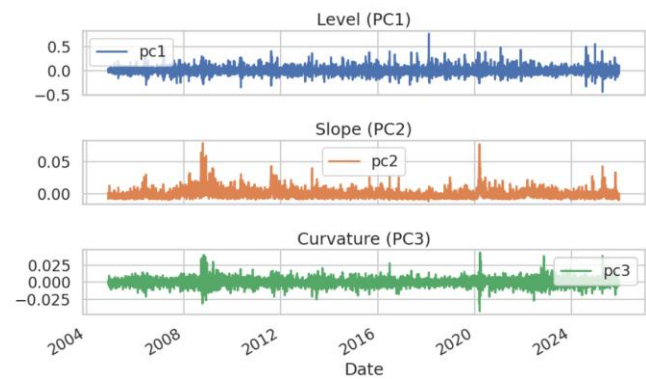


Figure 7. Time series of the first three principal components. The first component captures market-wide stress and exhibits pronounced spikes during crisis periods. The second primarily reflects cross-sectional dispersion across assets. The third behaves somewhat differently, showing short-lived curvature effects that tend to emerge around regime transitions.

Figures 6 and 7 illustrate the economic interpretation of the extracted components. The first principal component loads heavily on volatility-related variables and exhibits pronounced spikes during crisis periods, capturing market-wide stress analogous to a level factor. The second component reflects cross-sectional dispersion across assets, aligning with a slope-like factor associated with regime differentiation. The third component displays short-lived deviations around zero, consistent with curvature effects that emerge during regime transitions. Together, these components provide an economically meaningful decomposition that supports both regime identification and supervised stress prediction.

While PCA offers interpretability, computational efficiency, and effective noise reduction, it also has limitations. The method captures only linear relationships, and factor loadings may shift across market regimes, reducing stability over time. Moreover, principal components are statistical constructs and may lack direct economic meaning outside well-studied contexts such as fixed income. Despite these limitations, PCA provides a practical and transparent preprocessing step that

improves robustness when combined with rolling estimation and regime-aware modeling (Diebold and Li, 2006).

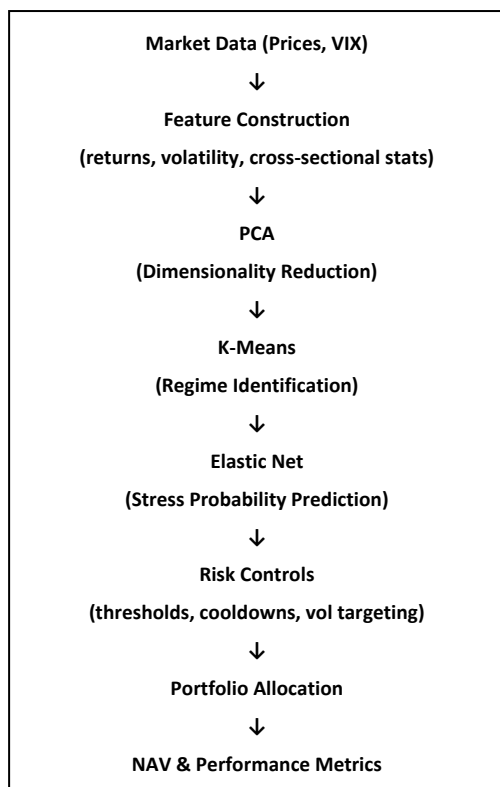


Figure 8. End-to-End Workflow of the proposed hybrid framework

Figure 8 summarizes the modeling pipeline used in this study. Raw market data are first transformed using Principal Component Analysis (PCA) to reduce dimensionality and suppress noise. The resulting latent factors are then used for regime identification via K-Means clustering, while Elastic Net regression generates forward-looking stress signals. These components are combined to inform dynamic portfolio allocation decisions within a time-series-aware framework.

4. Model Implementation and Validation

This section describes how the previously defined components are implemented within a time-series-aware evaluation framework intended to reflect realistic investment conditions. Hyperparameters are selected using rolling cross-validation and economic interpretability criteria rather than optimizing in-sample performance. This approach acknowledges the noisy, non-stationary, and regime-dependent nature of financial return data and prioritizes stability over time (Gu, Kelly, and Xiu, 2020; Harvey, Liu, and Zhu, 2016).

Dimensionality reduction is performed using Principal Component Analysis, with three components retained to

capture dominant systematic variation while avoiding reintroduction of idiosyncratic noise. Market regimes are identified using K-Means clustering with three clusters, corresponding to broad risk-off, neutral, and risk-on environments. These low-dimensional choices balance interpretability and flexibility and are consistent with empirical evidence showing improved out-of-sample stability for parsimonious regime structures (Kelly, Pruitt, and Su, 2019; Nystrup, Lindström, and Madsen, 2021).

Supervised stress prediction is implemented using Elastic Net regression with time-series cross-validation. Regularization parameters are selected from moderate ranges to avoid the instability of pure LASSO while maintaining sparsity and robustness under correlated predictors (Zou and Hastie, 2005; Feng, He, and Polson, 2020). Stress targets are defined using upper quantiles of volatility-related indicators to capture tail-risk events in a manner that adapts to changing market conditions (Adrian, Boyarchenko, and Giannone, 2019).

To reduce noise-driven regime switching, predicted stress probabilities are smoothed using rolling averages and combined with minimum holding period constraints. Portfolio construction incorporates volatility targeting to normalize risk exposure across regimes, improving comparability and drawdown control over time (Moreira and Muir, 2017). All allocation decisions are generated using walk-forward optimization, ensuring that model estimation and portfolio actions rely solely on information available at each point in time (López de Prado, 2018).

5. Performance Evaluation

Model performance is evaluated using a strict walk-forward framework designed to reflect realistic investment conditions. All signals, regime classifications, and portfolio allocations are generated sequentially using only information available at each point in time. This evaluation design mitigates look-ahead bias and aligns with best practices in financial machine learning (López de Prado, 2018; Harvey, Liu, and Zhu, 2016).

5.1 Predictive Performance

Predictive accuracy of the Elastic Net stress model is assessed using root mean squared error (RMSE). The hybrid framework achieves an RMSE of approximately 0.57, reflecting moderate predictive accuracy. While this level of accuracy may appear modest, it is consistent with

the low signal-to-noise nature of financial stress prediction, where even small timing improvements can have meaningful economic value (Gu, Kelly, and Xiu, 2020). Predictive accuracy is therefore evaluated as a supporting diagnostic rather than as the primary performance objective.

5.2 Portfolio-Level Performance

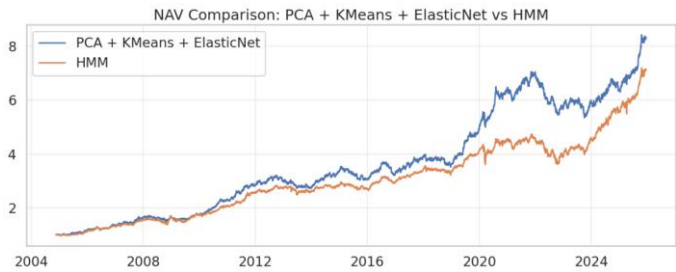


Figure 9. NAV Comparison

Figure 9 compares the walk-forward net asset value (NAV) of the proposed framework with a Hidden Markov Model (HMM) benchmark. While both strategies exhibit similar long-term volatility, the hybrid framework achieves higher terminal NAV and shallower drawdowns during major stress periods, including the Global Financial Crisis and the COVID-19 shock. These improvements are driven by earlier defensive positioning rather than increased leverage.

Table 1. Annualized Strategy Metrics (Walk-Forward)

Metric	Hybrid	HMM
Annualized Return	10.61%	9.78%
CAGR	10.40%	9.83%
Annualized Volatility	10.20%	10.28%
Sharpe Ratio	1.04	0.95
Maximum Drawdown	-24.26%	-23.76%
95% CVaR (daily)	-1.48%	-1.46%

Note: 95% conditional value-at-risk computed from daily return distribution

Table 2. Sharpe Ratio Comparison (Transaction-Cost Adjusted)

Metric	Hybrid	HMM
Sharpe Ratio (TC-adjusted)	1.03	0.95
Sharpe z-stat (Mommel adjustment)	N/A	0.61

Note: The Mommel Sharpe z-statistic is defined for pairwise comparison between strategies and is therefore reported only for the Hybrid vs. HMM comparison

Table 1 reports annualized performance metrics. The hybrid strategy delivers a slightly higher Sharpe ratio than the HMM benchmark, alongside comparable volatility and improved drawdown characteristics. To assess whether this difference is statistically meaningful, a Mommel-adjusted Sharpe ratio test is applied. The

resulting test statistic does not reject the null hypothesis of equal risk-adjusted performance at conventional confidence levels. CAGR is computed from the walk-forward NAV series and reported for descriptive purposes to summarize long-term capital growth. Downside tail risk, measured by daily 95% CVaR, is comparable across the two strategies, indicating that, in practical terms, performance differences arise primarily from risk timing and compounding. They do not appear to be driven by improved extreme-loss protection or tail hedging.

5.3 Interpretation and Robustness

The absence of strong statistical significance does not, in our view, invalidate the economic relevance of the results. Prior research shows that return-based tests often have low power in finite samples and that economically meaningful improvements frequently emerge through gradual compounding rather than sharp statistical breaks (Harvey, Liu, and Zhu, 2016). In this context, the observed outperformance reflects improved risk timing and regime adaptation rather than overfitting or excessive model complexity.

Overall, the performance evaluation suggests that the proposed framework enhances robustness and drawdown control while maintaining realistic turnover and volatility levels. Failure would manifest as persistently unstable allocations, excessive turnover, or materially worse drawdown behavior relative to the benchmark under identical evaluation conditions.

6. Conclusion

This paper presents a hybrid machine learning framework for investment strategy design that combines dimensionality reduction, unsupervised regime identification, and regularized stress prediction within a walk-forward evaluation setting. Rather than relying on complex architectures or aggressive optimization, the framework emphasizes stability, interpretability, and realistic validation.

Empirical results show that the hybrid strategy achieves higher terminal portfolio values and improved drawdown behavior relative to a Hidden Markov Model benchmark, although differences in risk-adjusted performance are not statistically decisive. This outcome is consistent with prior evidence that return-based tests often lack power in finite samples and that economically meaningful improvements may emerge gradually

through improved risk timing rather than sharp performance breaks.

From a practitioner's perspective, the most important result is not the marginal improvement in Sharpe ratios, but the stability of behavior across market regimes. Several alternative specifications were explored during development and ultimately rejected due to unstable allocations or excessive sensitivity to noise. The framework presented here reflects a conservative design choice, favoring robustness and interpretability over incremental gains in backtest performance

From a practical perspective, the results suggest that the primary contribution of machine learning in portfolio management may lie less in predictive accuracy and more in how models are structured, constrained, and evaluated. In our experiments, choices around regularization and validation had a comparable, and sometimes larger, impact on outcomes than the specific learning algorithms themselves. While the framework remains subject to the usual limitations of regime-based and data-driven approaches, it offers a transparent reference point for integrating machine learning techniques into investment processes without sacrificing economic interpretability or implementation discipline.

7. Model Limitations

Despite the use of regularization and time-series-aware validation, the proposed framework remains subject to several limitations common to machine learning applications in finance. The combination of dimensionality reduction, clustering, and supervised learning introduces multiple modeling choices, including the number of principal components, regime partitions, and regularization strength. If not carefully validated using strict walk-forward procedures, such flexibility can increase model selection risk and amplify sensitivity to noise rather than persistent structure (Harvey, Liu, and Zhu, 2016; Gu, Kelly, and Xiu, 2020).

The regime identification component relies on K-Means clustering, which assumes that historical regime structures are sufficiently stable to remain informative out of sample. Structural breaks, policy shifts, or extreme

market events may alter regime dynamics in ways that historical clustering cannot anticipate, reducing effectiveness during periods of rapid transition (Nystrup, Lindström, and Madsen, 2021).

Finally, while Elastic Net regression provides interpretable coefficients, combining it with PCA and unsupervised regime identification introduces trade-offs in transparency. Principal components are statistical constructs that may lack direct economic interpretation, and regime labels are not explicitly tied to macroeconomic drivers. As a result, attribution of performance improvements to specific economic mechanisms remains imperfect (Feng, He, and Polson, 2019; Kelly, Pruitt, and Su, 2019).

8. Future Research

Several extensions may further enhance the framework while preserving its emphasis on robustness and interpretability. One avenue is the use of adaptive regularization or regime-dependent hyperparameters, allowing model complexity to vary across market environments. Such approaches may improve stability during structural shifts while maintaining disciplined shrinkage.

Another direction involves exploring alternative regime-detection methods, including probabilistic or density-based clustering techniques such as Hidden Markov Models or Gaussian Mixture Models. These methods may better capture regime transitions and overlapping states, particularly during periods of market stress, though often at the cost of increased model complexity (Nystrup, Lindström, and Madsen, 2021).

Finally, incorporating broader information sets, including macroeconomic indicators or cross-asset signals, may improve the economic interpretability and forward-looking capacity of regime identification. While nonlinear factor extraction methods and deep learning architectures offer additional flexibility, their use would require careful consideration of transparency, estimation risk, and validation, which remain central concerns in financial machine learning research (Gu, Kelly, and Xiu, 2020; López de Prado, 2018).

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